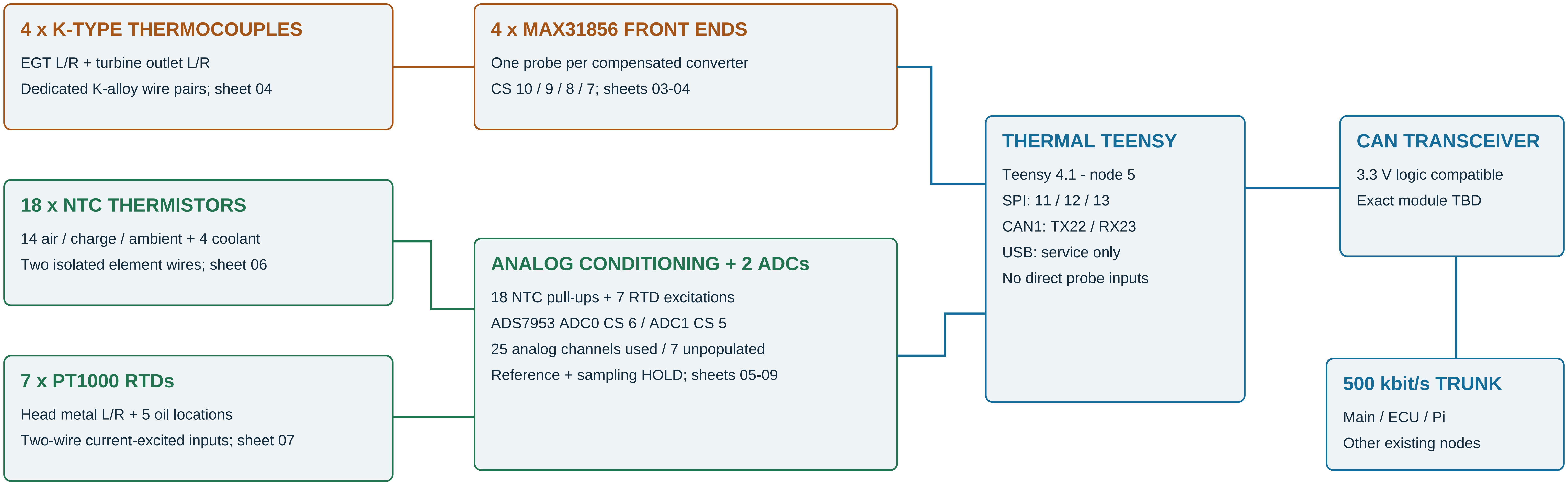
Whole-subsystem wiring overview

29 active sensors / one dedicated Teensy 4.1 / six SPI front ends / CAN to main controller and HUD



PROTECTED POWER / SHEET 02

Fused switched supply -> protected DC/DC
5 V VIN; separate regulated peripheral rails
Power return and shield bonds are distinct

INDEPENDENT ECU SENSORS RETAINED

ECU coolant + IAT + oil temperature stay wired
Do not parallel thermal and ECU sensor inputs
No ECU replacement-input wiring approved here

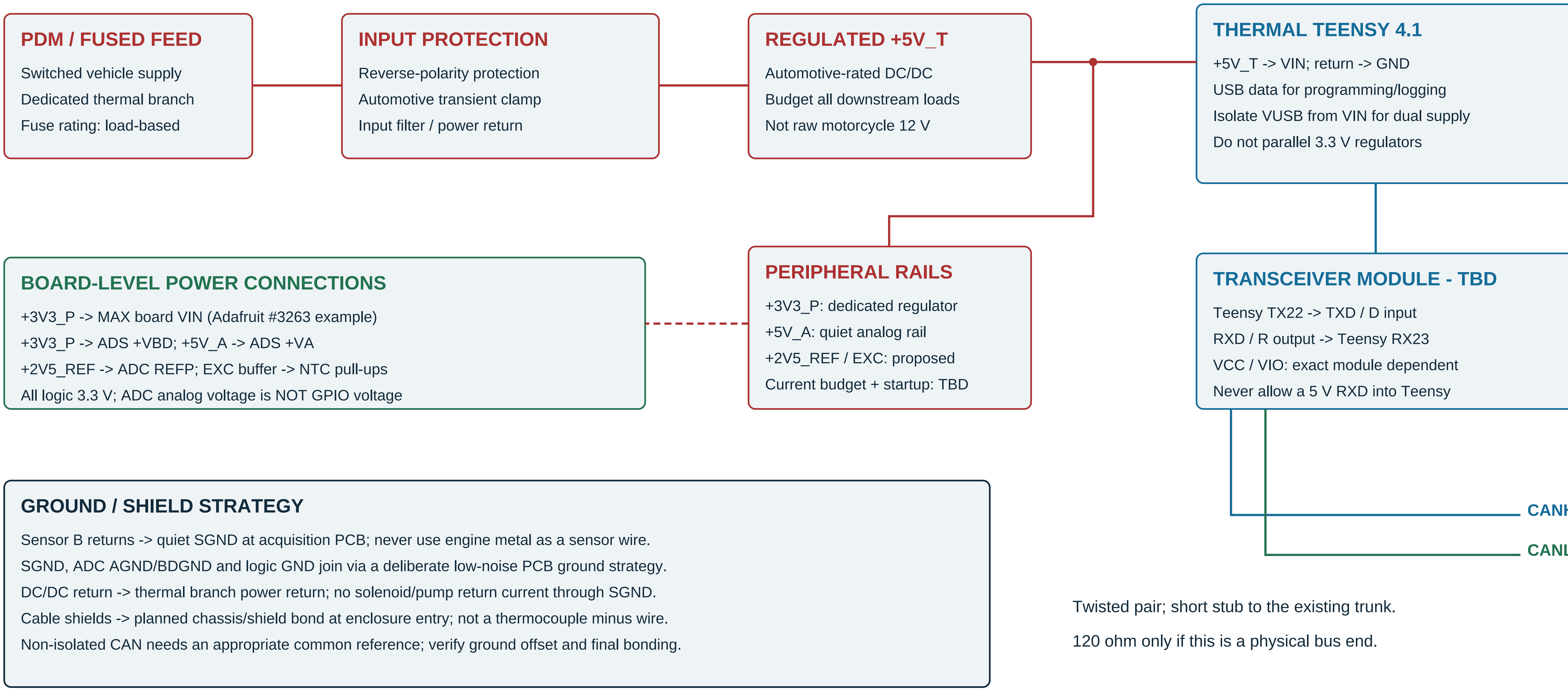
DRAWING KEY

Solid named net = defined signal assignment
Dashed = proposed / unresolved implementation
Dot = connection; crossing without dot = no join

Scope: complete functional harness map, not a purchased-board cavity map or a finished analog PCB schematic.

Power distribution, returns and CAN

Functional connections; fuse, DC/DC, transceiver and connector part numbers remain to be selected

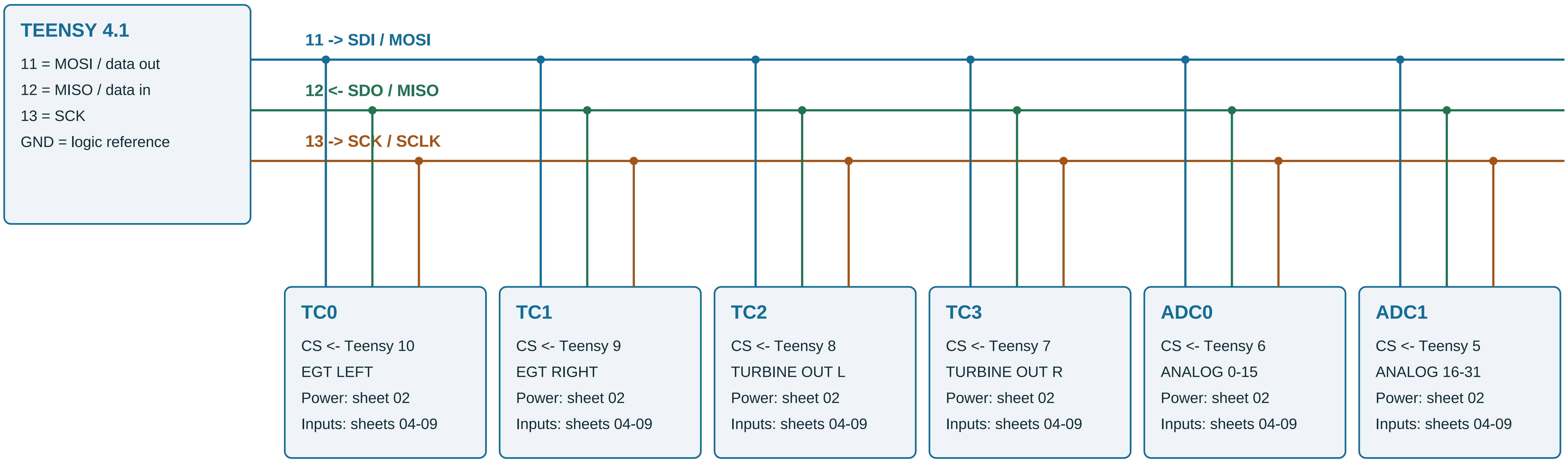


Source [P1]. Proposed separate peripheral regulator avoids assuming spare current capacity on the Teensy regulator.

Do not back-power unpowered boards through SPI. Confirm rail sequencing and any required isolation/series protection.

SPI and chip-select wiring

Every module shares MOSI, MISO and SCK; each has its own active-low chip select



MODULE PAD NAME CROSSWALK

Module	MOSI net 11	MISO net 12	Clock net 13	Chip select	Unused / power notes
MAX31856 board	SDI	SDO	SCK	CS (per above)	FLT, DRDY, 3Vo not connected
ADS7953 board	SDI	SDO	SCLK	CS (per above)	GPIOs unused; configure safe state

Proposed: a 10 kOhm pull-up on each CS to its 3.3 V logic rail. Check for existing pull-ups before adding.

Keep SPI inside the acquisition enclosure; verify unselected SDO is high impedance. No breadboard-length vehicle SPI.

Source: repository driver headers; Adafruit shared-SPI guidance [A1]. Crossings without dots are not connected.

Thermocouple harnesses and breakout boards

Four independent K-type pairs; proposed example breakout: Adafruit MAX31856 #3263, not a generic pin-order guarantee

Probe	Pairs	Breakout	Connections
T01 / EGT_LEFT left exhaust pre-turbine K-type probe; insulated junction preferred	K+ -> T+ K- -> T-	**TC0 / MAX31856** T+ = positive / T- = negative CS = Teensy 10	VIN <- +3V3_P; GND <- logic GND SDI 11 / SDO 12 / SCK 13 3Vo, FLT and DRDY: leave unconnected
T02 / EGT_RIGHT right exhaust pre-turbine K-type probe; insulated junction preferred	K+ -> T+ K- -> T-	**TC1 / MAX31856** T+ = positive / T- = negative CS = Teensy 9	VIN <- +3V3_P; GND <- logic GND SDI 11 / SDO 12 / SCK 13 3Vo, FLT and DRDY: leave unconnected
T03 / TURBINE_OUT_LEFT left turbine outlet K-type probe; insulated junction preferred	K+ -> T+ K- -> T-	**TC2 / MAX31856** T+ = positive / T- = negative CS = Teensy 8	VIN <- +3V3_P; GND <- logic GND SDI 11 / SDO 12 / SCK 13 3Vo, FLT and DRDY: leave unconnected
T04 / TURBINE_OUT_RIGHT right turbine outlet K-type probe; insulated junction preferred	K+ -> T+ K- -> T-	**TC3 / MAX31856** T+ = positive / T- = negative CS = Teensy 7	VIN <- +3V3_P; GND <- logic GND SDI 11 / SDO 12 / SCK 13 3Vo, FLT and DRDY: leave unconnected

THERMOCOUPLE ROUTING IS NOT ORDINARY COPPER SENSOR WIRING

Keep K-alloy extension cable and matched polarity connectors through to the cold-junction terminals. Do not tie T- to chassis or SGND. Confirm grounded-junction compatibility and input common-mode limits before using a grounded probe.

Keep converter boards away from heat gradients. Shared logic ground does not make these inputs galvanically isolated. Source [A1].

Analog front-end board connections

ADC0 and ADC1 repeat this functional circuit; physical package / breakout connector pin numbers are not selected

16 CONDITIONED INPUTS

CH0 ... CH15 from sheets 08/09
NTC / RTD circuits sheets 06/07
Clamp exposed harness faults
RC / source settling: design TBD

ADS7953 x 2

+VA <- +5V_A (proposed)
+VBD <- +3V3_P
REFP <- +2V5_REF (proposed)
REFM / AINM -> AGND
AGND / BDGND -> local ground
SDI / SDO / SCLK / CS: sheet 03

REFERENCE / EXCITATION BLOCK - HOLD

Precision 2.5 V reference + local bypassing
Separate buffered 2.5 V NTC excitation, ratio-tracked to REF
Driver/conversion calibration must match this proposal

MUX OUTPUT PATH MUST BE PRESENT

ADS MXO -> suitable unity buffer/filter -> ADS AINP
AINM -> analog return (single-ended measurement)
Select buffer and settling network for the actual source
No open MXO/AINP path; inspect any breakout schematic

LOCAL PCB DETAILS STILL REQUIRED

Local supply and reference decoupling; validated ADC input protection
RTD current-source compliance / stability and NTC excitation loading
ADC input RC, buffer stability and multiplexer settling across channel steps
Unpopulated channels: terminate/park per PCB design; do not add phantom sensors
No ADC channel connects directly to a Teensy analog pin

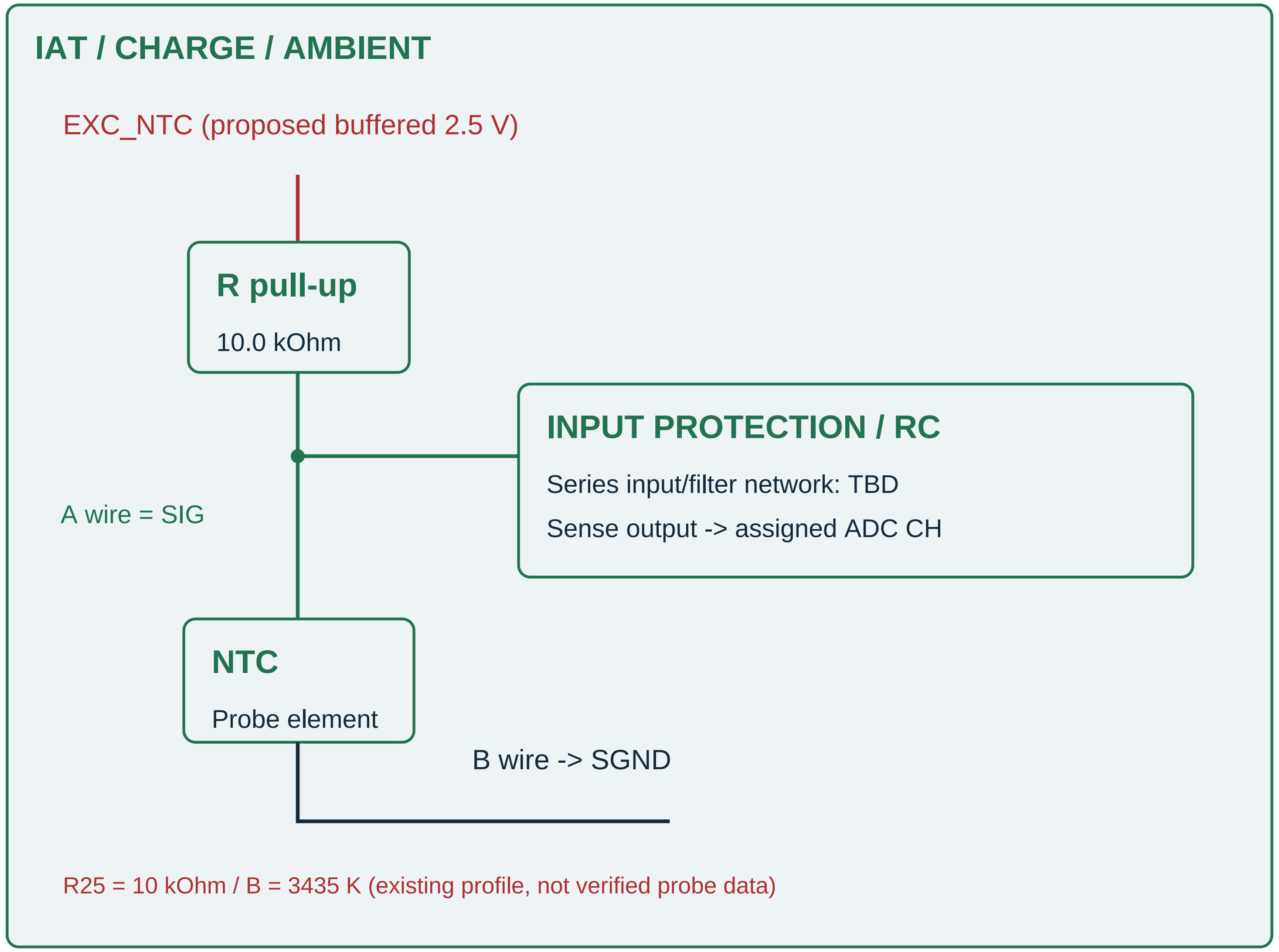
HOLD: CURRENT FIRMWARE IS NOT YET COMPATIBLE WITH A RELEASED ANALOG DESIGN

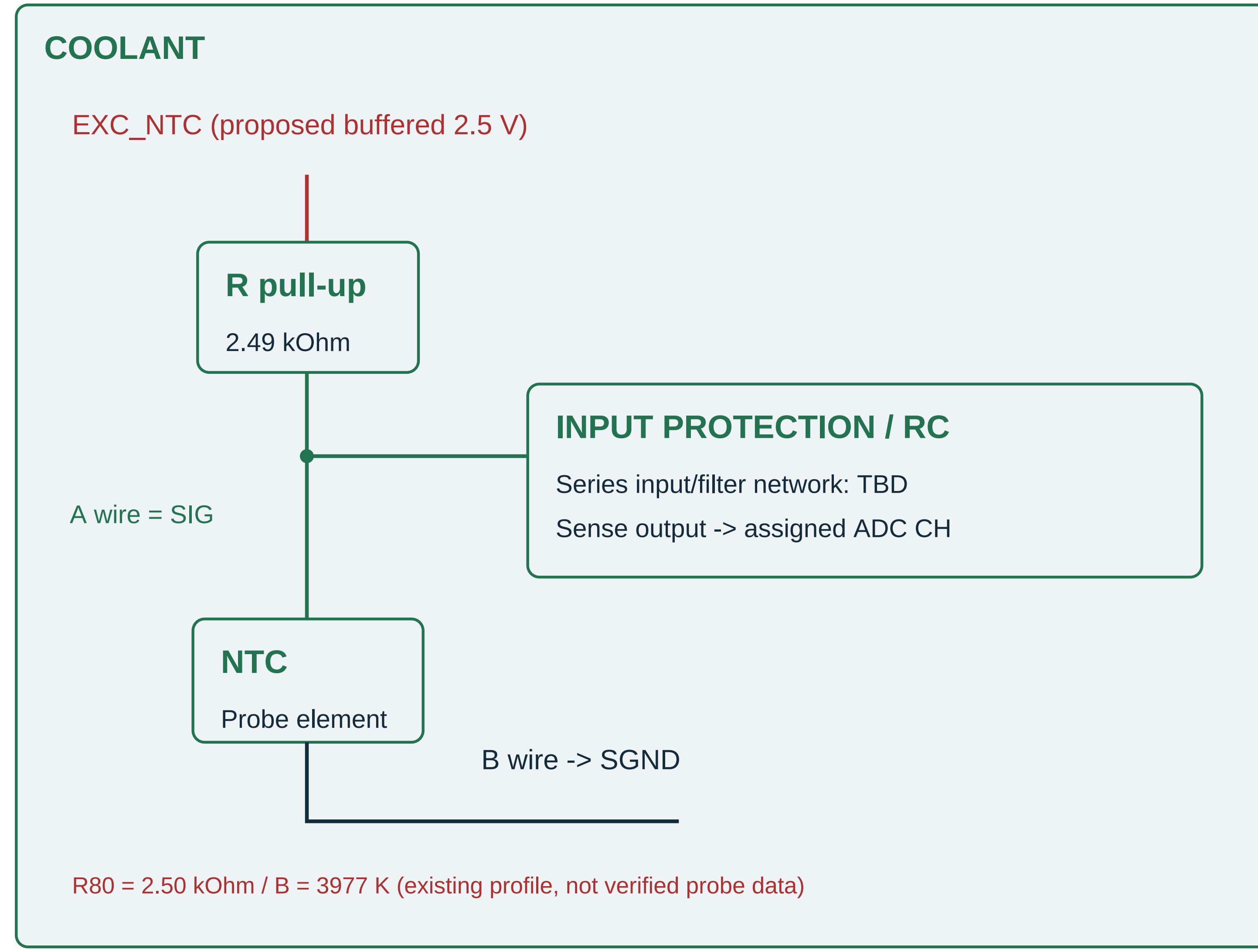
ADS7953 REF permits 2.0-3.0 V, not 3.3 V. Current RTD conversion assumes a 3.3 V full scale. The two-frame read also needs N+2 pipeline verification. Resolve range, excitation ratio, priming and channel tags before trusting temperatures.

Proposed implementation, not a selected ADC breakout. Source [T1] plus repository analog_adc_driver.cpp / sensor_conversion.cpp.

NTC conditioning - repeat for 18 channels

14 air/charge/ambient channels use the IAT profile; four coolant channels use their separate profile





THE PULL-UP CIRCUIT IS REQUIRED; A BREAKOUT ADC DOES NOT SUPPLY IT

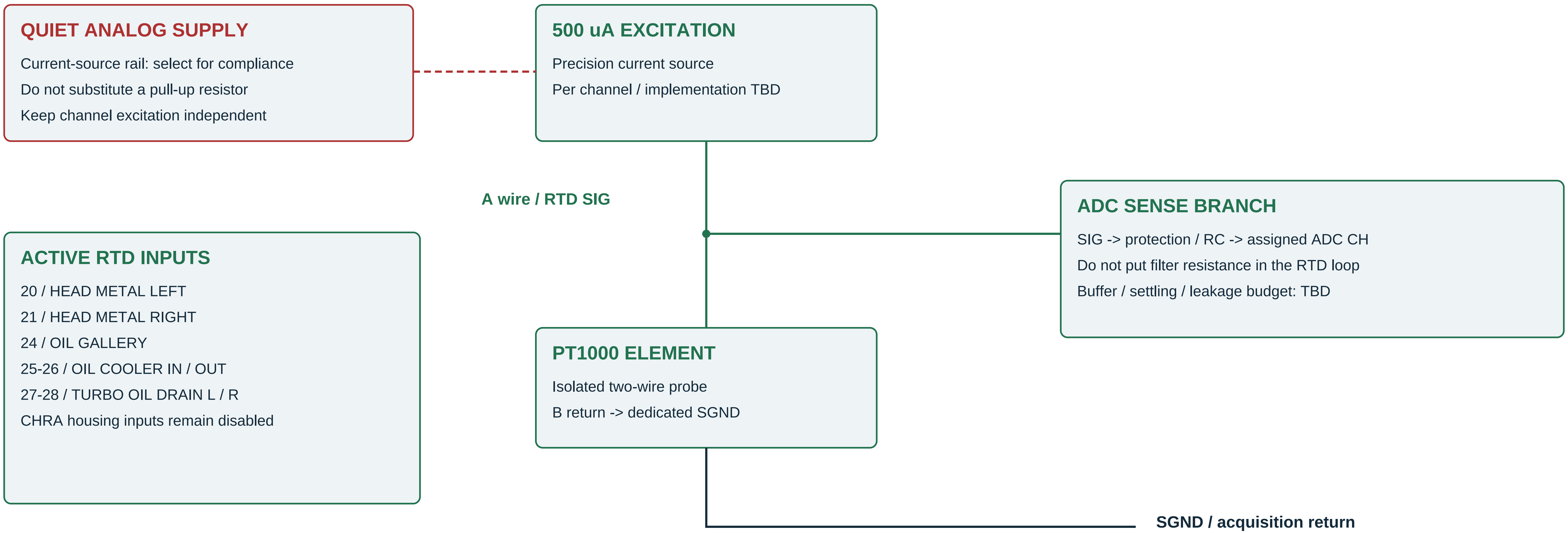
Each NTC has its own pull-up and dedicated signal/return pair. Excitation must track ADC full scale for the existing resistance-ratio formula. A 3.3 V pull-up with a 2.5 V full scale is not interchangeable.

Do not assume a common automotive coolant sender matches 2.50 kOhm at 80 C. Obtain actual resistance tables.

Specify probe temperature rating, sealing, thread, response time and WMI-fluid compatibility before ordering. No ECU sensor sharing.

PT1000 conditioning - repeat for seven channels

A 500 uA precision excitation is assumed by the current conversion code; the actual conditioning board is not designed yet



TWO-WIRE LIMITATION AND CONVERSION HOLD

Lead resistance is included in this measurement. Three/four-wire compensation is not implemented. Validate self-heating, current accuracy and full temperature conversion; the existing linear RTD model is an approximation and must be recalibrated for the chosen ADC range.

Keep both probe conductors isolated from chassis. Open/short protection must not damage the current source or ADC.

A MAX31865 RTD breakout is not a drop-in replacement for this ADS7953/current-source architecture; firmware would change.

ADC0 channel-by-channel harness

ADS7953 device 0 / CS = Teensy 6 / logical analog channels 0-15

Each row is one complete two-wire sensor circuit. A/B are proposed harness net labels, NOT manufacturer connector cavity numbers.

ADC pin	Logical	ID	Sensor / reserved name	Conditioning	Signal wire path	Return wire path
CH0	0	5	COMP_IN_LEFT	IAT / 10k	T05-A -> SIG -> CH0	T05-B -> SGND
CH1	1	6	COMP_IN_RIGHT	IAT / 10k	T06-A -> SIG -> CH1	T06-B -> SGND
CH2	2	7	COMP_OUT_LEFT	IAT / 10k	T07-A -> SIG -> CH2	T07-B -> SGND
CH3	3	8	COMP_OUT_RIGHT	IAT / 10k	T08-A -> SIG -> CH3	T08-B -> SGND
CH4	4	9	IC_IN_LEFT	IAT / 10k	T09-A -> SIG -> CH4	T09-B -> SGND
CH5	5	10	IC_IN_RIGHT	IAT / 10k	T10-A -> SIG -> CH5	T10-B -> SGND
CH6	6	11	IC_OUT_LEFT	IAT / 10k	T11-A -> SIG -> CH6	T11-B -> SGND
CH7	7	12	IC_OUT_RIGHT	IAT / 10k	T12-A -> SIG -> CH7	T12-B -> SGND
CH8	8	13	PRE_WMI	IAT / 10k	T13-A -> SIG -> CH8	T13-B -> SGND
CH9	9	14	POST_WMI	IAT / 10k	T14-A -> SIG -> CH9	T14-B -> SGND
CH10	10	15	PLENUM_IAT	IAT / 10k	T15-A -> SIG -> CH10	T15-B -> SGND
CH11	11	16	RUNNER_IAT_LEFT	IAT / 10k	T16-A -> SIG -> CH11	T16-B -> SGND
CH12	12	17	RUNNER_IAT_RIGHT	IAT / 10k	T17-A -> SIG -> CH12	T17-B -> SGND
CH13	13	18	HEAD_COOLANT_LEFT	CLT / 2.49k	T18-A -> SIG -> CH13	T18-B -> SGND
CH14	14	19	HEAD_COOLANT_RIGHT	CLT / 2.49k	T19-A -> SIG -> CH14	T19-B -> SGND
CH15	15	20	HEAD_METAL_LEFT	RTD / 500 uA	T20-A -> SIG -> CH15	T20-B -> SGND

All input filtering/protection is on the acquisition side of the harness connector. See sheets 05-07 for circuit topology.

ID32 has no physical input assignment. The stable ID namespace is not the same thing as ADC channel numbering.

ADC1 channel-by-channel harness

ADS7953 device 1 / CS = Teensy 5 / logical analog channels 16-31

Each row is one complete two-wire sensor circuit. A/B are proposed harness net labels, NOT manufacturer connector cavity numbers.

ADC pin	Logical	ID	Sensor / reserved name	Conditioning	Signal wire path	Return wire path
CH0	16	21	HEAD_METAL_RIGHT	RTD / 500 uA	T21-A -> SIG -> CH0	T21-B -> SGND
CH1	17	22	RAD_IN	CLT / 2.49k	T22-A -> SIG -> CH1	T22-B -> SGND
CH2	18	23	RAD_OUT	CLT / 2.49k	T23-A -> SIG -> CH2	T23-B -> SGND
CH3	19	24	OIL_GALLERY	RTD / 500 uA	T24-A -> SIG -> CH3	T24-B -> SGND
CH4	20	25	OIL_COOLER_IN	RTD / 500 uA	T25-A -> SIG -> CH4	T25-B -> SGND
CH5	21	26	OIL_COOLER_OUT	RTD / 500 uA	T26-A -> SIG -> CH5	T26-B -> SGND
CH6	22	27	TURBO_OIL_DRAIN_LEFT	RTD / 500 uA	T27-A -> SIG -> CH6	T27-B -> SGND
CH7	23	28	TURBO_OIL_DRAIN_RIGHT	RTD / 500 uA	T28-A -> SIG -> CH7	T28-B -> SGND
CH8	24	29	AMBIENT_AIR	IAT / 10k	T29-A -> SIG -> CH8	T29-B -> SGND
CH9	25	30	CHRA_TEMP_LEFT	NOT FITTED	No sensor lead	No sensor return
CH10	26	31	CHRA_TEMP_RIGHT	NOT FITTED	No sensor lead	No sensor return
CH11	27	-	UNALLOCATED SPARE	NOT FITTED	No sensor lead	No sensor return
CH12	28	-	UNALLOCATED SPARE	NOT FITTED	No sensor lead	No sensor return
CH13	29	-	UNALLOCATED SPARE	NOT FITTED	No sensor lead	No sensor return
CH14	30	-	UNALLOCATED SPARE	NOT FITTED	No sensor lead	No sensor return
CH15	31	-	UNALLOCATED SPARE	NOT FITTED	No sensor lead	No sensor return

All input filtering/protection is on the acquisition side of the harness connector. See sheets 05-07 for circuit topology.

ID32 has no physical input assignment. The stable ID namespace is not the same thing as ADC channel numbering.

Build boundary, connector schedule and checks

Close these items before turning the planning pack into a fabrication-ready schematic

Connection group	Defined now	Still required
T01-T04 / 4 K pairs	TC polarity, probe role, converter CS	K connector family, grounded/ungrounded probe part
T05-T29 / 25 A/B pairs	Signal role, sensor profile, ADC channel	Sealed connector cavities, wire gauge/length, probe PN
Power / thermal branch	VIN 5 V, regulated peripheral rails, return	Fuse/DC/DC/transient devices and measured load budget
CAN interface	TX22 to TXD; RXD to RX23; H/L trunk	Exact 3.3 V-compatible module and termination location
Analog acquisition PCB	Channel allocation and required circuit blocks	Reference, excitation, buffer, filter, protection schematic
Firmware commissioning	Source sensor IDs and baseline profiles	ADC pipeline/range repair and measured calibration

BENCH CHECKLIST

1. Check power polarity, shorts and USB/VIN isolation before power.
2. Verify rails / logic levels with sensors disconnected.
3. Confirm each probe label by substituting one known input at a time.
4. Test resistor/RTD standards and K-type simulator points.
5. Validate open, short, crossed lead, stale node and power-loss faults.
6. Compare raw codes to expected voltage and confirm channel identity.
7. Confirm main-controller thermal fallback with the Pi unplugged.

IMPORTANT FINDINGS / NO FIRMWARE EDIT IN THIS PACK

H1: 3.3 V conversion assumption versus ADS reference/range.

H2: manual-mode channel return is N+2; current read is two frames.

H3: ADS board and excitation/protection circuits are not selected.

H4: exact sensor curves, connectors and shield bonds are unverified.

H5: keep independent ECU coolant, IAT and oil-temperature inputs.

Do not order an arbitrary ADS7953 module and assume it includes NTC pull-ups, RTD current sources, buffer or a valid reference.

[P1] PJRC Teensy 4.1 pin card; TX22 / RX23 https://www.pjrc.com/teensy/card11a_rev4_web.pdf

[A1] Adafruit MAX31856 breakout pinouts https://learn.adafruit.com/adafruit-max31856-thermocouple-amplifier/pinouts

[T1] TI ADS7953 datasheet, SLAS605C; reference / MXO / Fig. 51 https://www.ti.com/lit/ds/symlink/ads7953.pdf

Repository: config/thermal_system.json and arduino/teensy41/albatross_thermal_node/*; source hash on every sheet.